

AI and ML at work

- ① Health care:
- ② Finance: ML model $\rightarrow$ 100 feature
location, Amount, < 50
- ③ E-commerce: 357.
- ④ Entertainment: Netflix
- ⑤ Agriculture: John Deere 98%
- ⑥ Language: Google translate neural networks
- ⑦ Self driving car: waymo, Tesla
- ⑧ Generative AI:

modern AI - end-to-end driving!

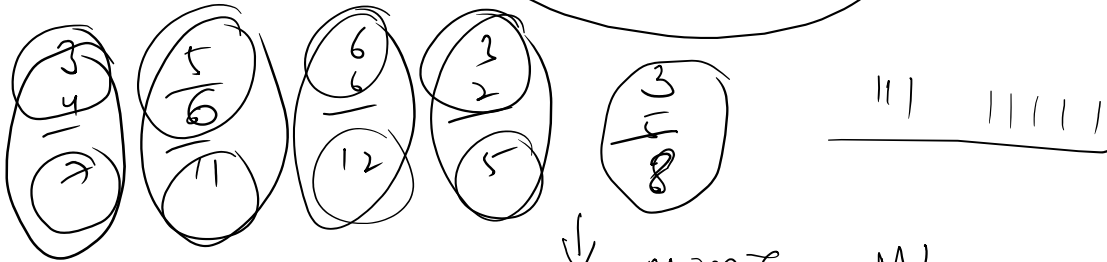
The old way (2005-2011): Had coded rules

middle way (2011-20): ① ② ③

modern way: (2020-till)



ML $\Rightarrow$ (ML)



$\downarrow$ mimic $\rightarrow$ ML

ML



Assertion $\rightarrow$

9



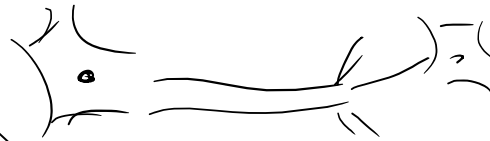
1586

data

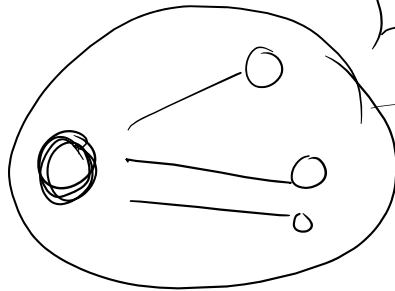
"img Audio, video" $\rightarrow$

(data)

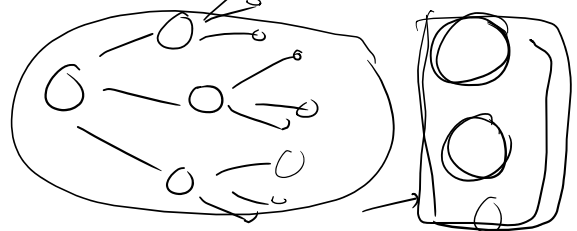
neuron



2+3



2.3 + 5.6.7.2 + 36



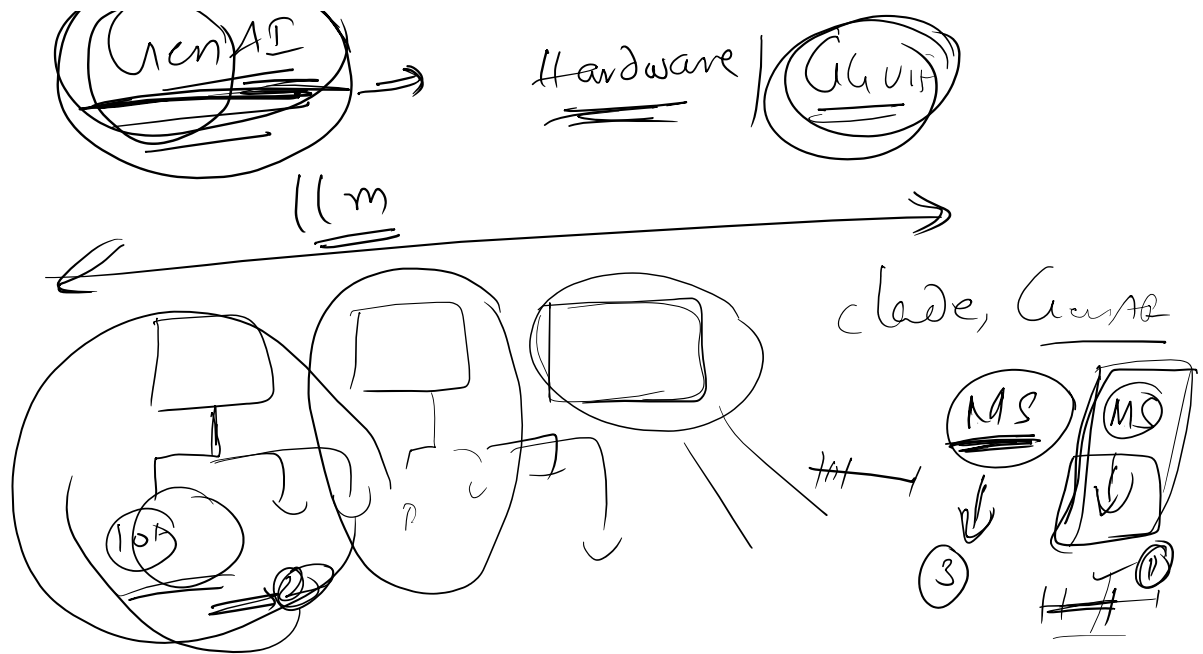
mimic

Deep learning img, Audio, video

Rule $\rightarrow$ ML $\rightarrow$ DL $\rightarrow$ 2017 $\rightarrow$ predictive

CNN

Hardware | GPU



$$\frac{3}{4} \quad \frac{3}{6} \quad \frac{7}{15}$$

$$\frac{2}{4} \Rightarrow$$

$$y = x + y$$

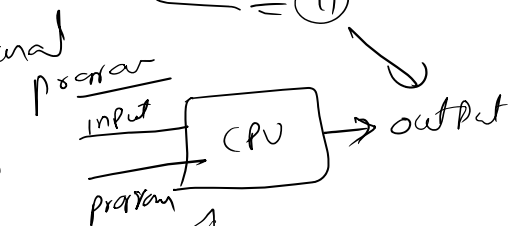
$$f(x) = x_1 + x_2$$

$$= 3 + 8$$

$$= 11$$

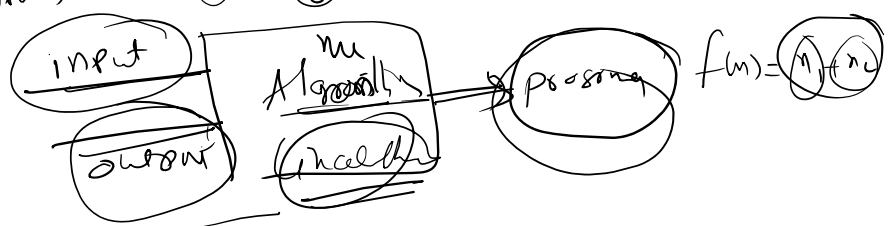
a = Enter number
 b = Enter number
 c = a + b
 print(c)

program
 function
 traditional



$$\frac{4}{7} \quad \frac{3}{6} \quad \frac{4}{12}$$

ML



AI

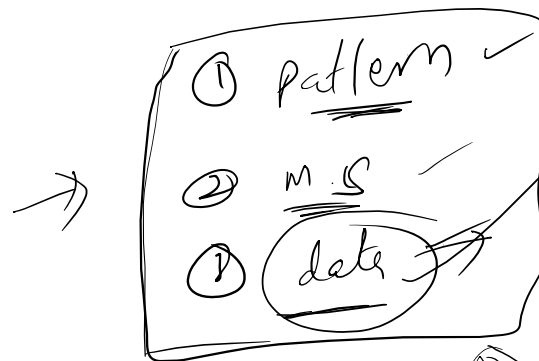
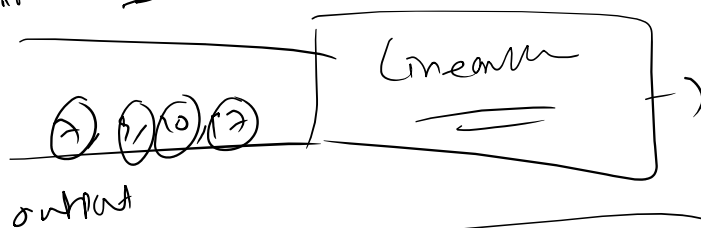
↳ Machine Learning (ML)

↳ Deep Learning (DL)

↳ Generative AI (GenAI)

↳ LLMs
(chatgpt, Claude, etc)

input [3, 4] (4, 5), (1, 1), (5, 0)



ML Algorithms classified into two categories

Supervised learning

Unsupervised learning

you train labeled data →

Input	Label
"win \$1,000 now"	Spam
"meeting at 7pm"	not spam

'meeting at 7pm' → not spam
'cheap kids lunch' → spam

 ?

predicting house price → size location price

Recloning have ionate $\rightarrow$ 0-3

U.L

10,000 $\rightarrow$

Group A: young shoppers when they are

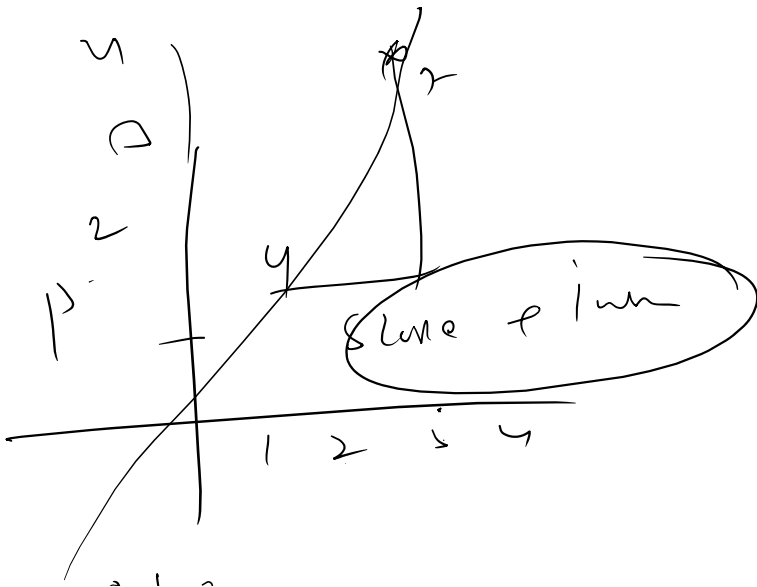
Ques B :- parent supply cross the

Ques: Senior customer buying medium

Analysis τC

$$y = \textcircled{m}x + c$$

$$y = \frac{m}{m_L} + m_L \frac{n_L}{C}$$



x	y
-----	-----

0 1 2 3 4

1, 3, 5, 7, 9 ?

data
observe

pattern → //

x	y
0	1
1	3
2	5
3	7
4	9
5	?

$x \in \text{input} = [0, 1, 2, 3, 4]$

$y(\text{output}) = [1, 3, 5, 7, 9]$

algorithm Fit () → fun

0.99 3.02 21.00 16.58, 33.0. ? - -

(x, y)

(x, y, c)

→

what makes it difficult

① uncertainty →

② dimension

min max

20

20

20

20

(10)

20

[1, 2] [2, 3] [4, 5]

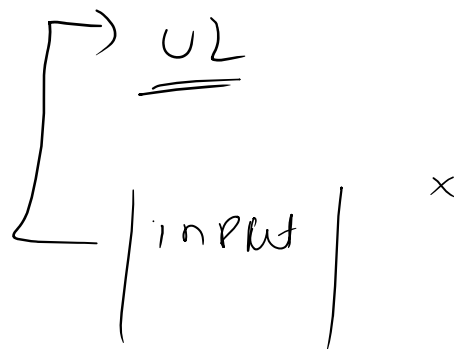
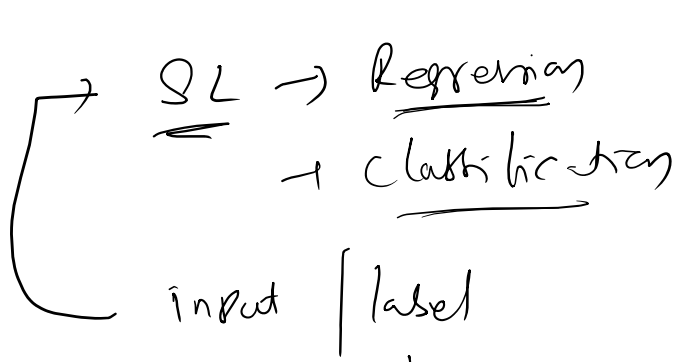
3

5

9

→ Q1 → Regression

→ U2



input / label $\leftarrow$ Continuous (Regression)

100k

min	max
20	36
21	31
22	30
40	35

Regression

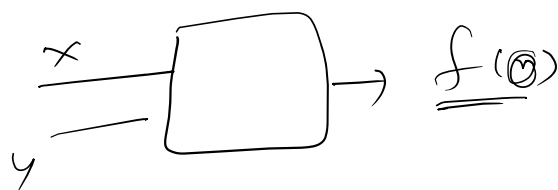
classification

SL

classification

Salary	debuter	city	loan
-	-	-	(yes)
-	-	-	(no)

(3, 4) (4, 5) (3, 2) (5, 3) features
 7 9 5 8 (input) label



x_1	x_2	y
3	4	7
4	5	9
2	2	5
5	3	8

$$f(\text{features}) = x_1 + x_2$$

$$f(x_1, x_2) = x_1 + x_2$$

$$f(6, 2) = 6 + 2 = 8$$

$$\left[\begin{array}{c|c|c} 1 & 4 & 5 \\ 5 & 3 & 8 \end{array} \right]$$

$$6 \quad 2 \quad 8$$

prediction function

Size	Shape	Color	name
small	circle	Red	apple
small	circle	yellow	banana

classification

$$f(\text{features}) = \text{name}$$

$$f(\text{size, shape, color}) = \text{name}$$

$$y = f(x) = x_1 + x_2$$

100 Sample

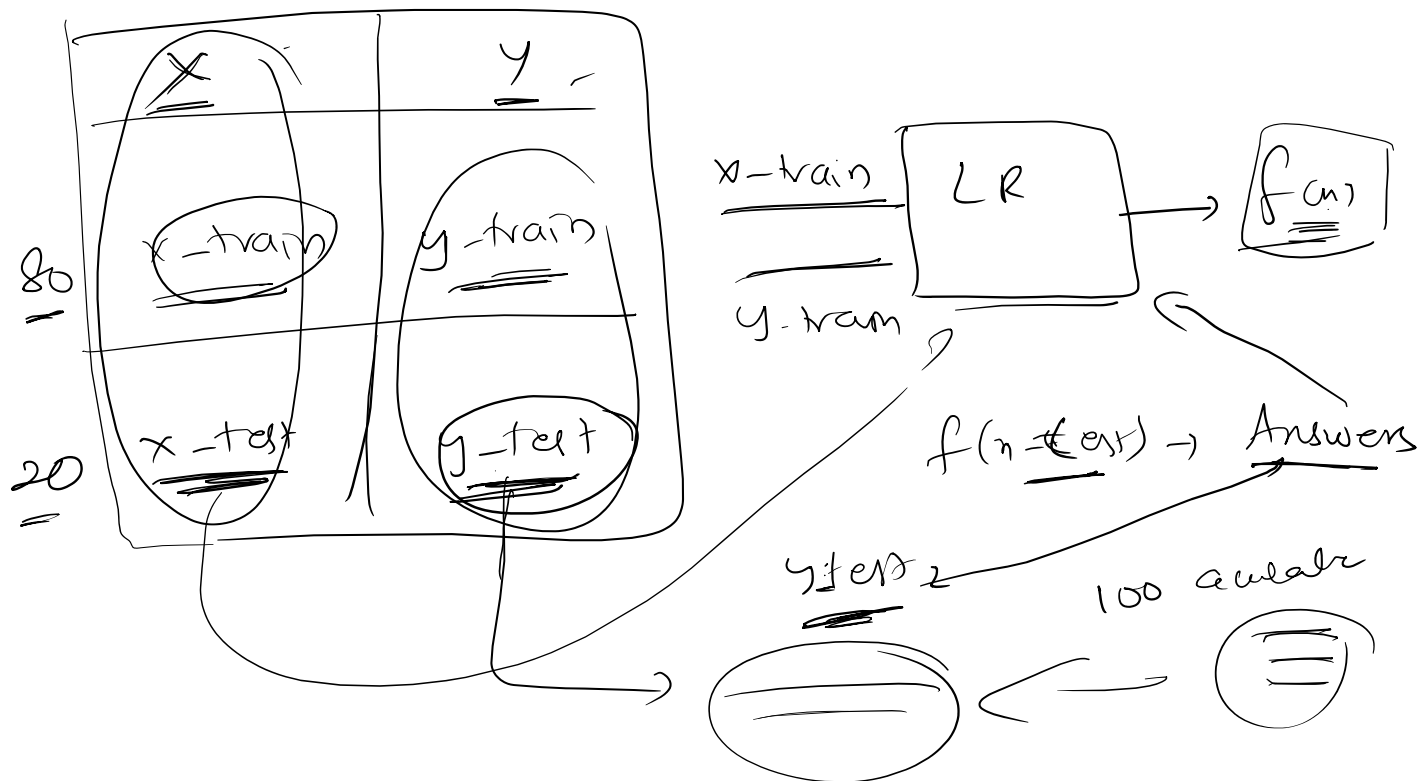
features					label
x_1	x_2	x_3	x_4		y
1	2	3	4		-
10	2.5	1.5	4		-

80 Sample

20

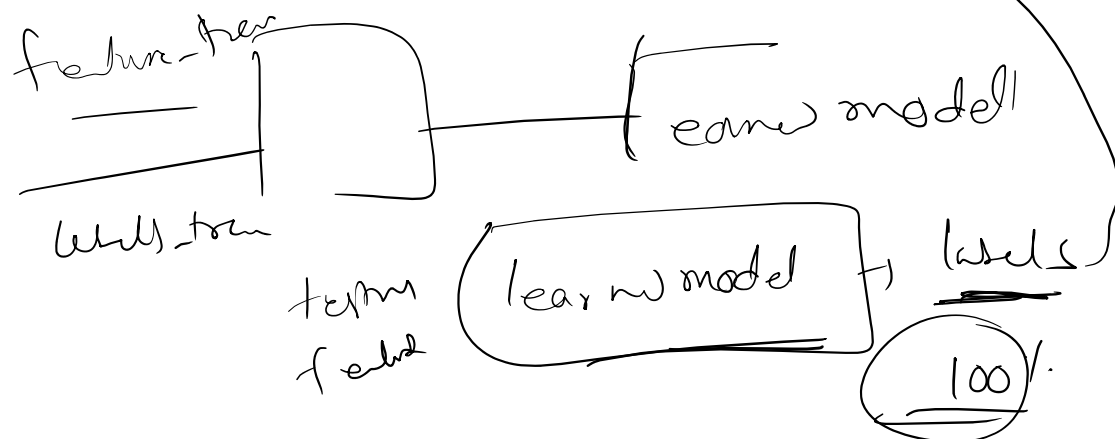
training

testing



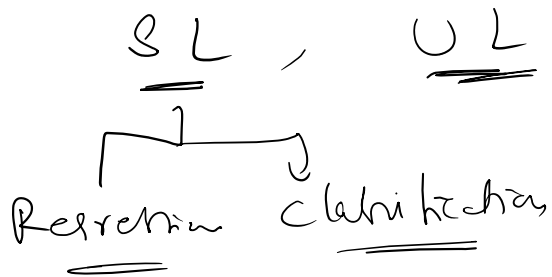
data $\rightarrow$

features	label
train	train
<u>test</u>	<u>test</u>

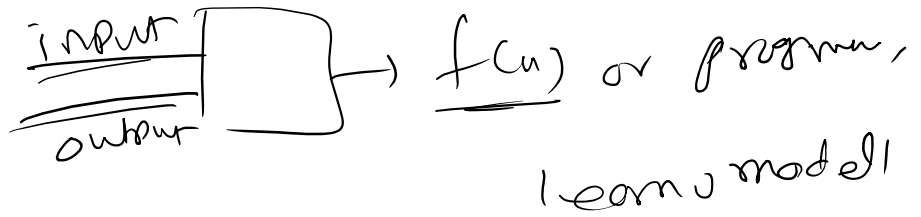
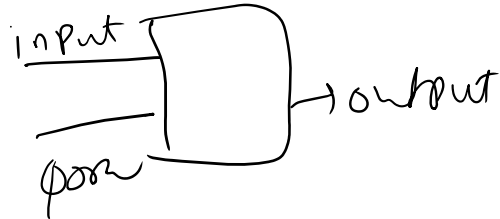


AI, ML, DL ✓

where we implement $\rightarrow$ ML



Group



So

20

	<u>X</u>	<u>Y</u>
	x-train	y-train
	<u>x-test</u>	<u>y-test</u>

